

Initial test of ED to LEO satellite transmission Visibility, Collision and Doppler effect

I. INTRODUCTION

THIS article presents an initial evaluation of End Device (ED) transmissions to LEO satellites. It aims to identify aspects that can be further explored in the development of a final paper. This article is divided into five sections. Section I presents this briefly introduction, Section II describes the test parameters that can be configured, Section III presents the evaluated parameters, Section IV shows the final results, and Section V presents the considerations.

II. CONFIGURABLE PARAMETERS

A. Simulation time interval

The simulation time window is defined by an initial time and a final time, representing the period over which the transmission from End Devices (EDs) to the LEO satellites are evaluated.

B. Satellite

A satellite constellation is chosen to be evaluated in the simulation. A minimum elevation angle between the satellite and the EDs is also configured.

C. End devices configuration

The ED nodes have five main attributes: the total number of devices, their distribution (or position), the transmission policy, the ED transmission power and the number of transmissions per ED.

D. LoRa configuration

The LoRa modulation has the following parameters: the Spreading Factor (SF), the Bandwidth (B) in Hz, the Low Data Rate Optimization (LDRO) mechanism, the packet length (L) in bytes, the communication frequency (satellite and nodes) in MHz.

III. EVALUATED PARAMETERS

A. PDR evaluation

The PDR of ED transmissions are evaluated using four main parameters:

- **Collision:** a packet collision occurs when, due to their transmission instant and transmission times (Time on Air, ToA), two transmitted packets overlap partially or completely in time, simultaneously occupying the same physical communication medium.
- **Visibility:** a transmission is considered visible when the elevation angle between the satellite and the end device (ED) is above a threshold determined as the minimum to established the communication (ex. 5 °).¹
- **Doppler effect:** a Doppler error occurs when the frequency shift caused by the Doppler effect results in an incorrect decoded symbol. This error could occurs in two demodulation moments: on the synchronization, when the static frequency shift exceeds a threshold, (for instance $B/4$ for $SF=\{7-9\}$), or during payload demodulation, when the accumulated frequency shift causes the FFT energy peak of a payload symbol to cross the threshold corresponding to the frequency spacing between adjacent symbols.
- **Successfully transmission:** a transmission is considered successful when a packet is transmitted and correctly received, with no occurrence of collision, visibility, or Doppler errors. Then the final PDR can be obtained thought the following equation:

$$PDR (\%) = \frac{N_S}{N_T} \cdot 100 \quad (1)$$

Where N_T is the total number of transmissions and N_S the total number of sucessfully transmissions.

¹It should be noted that a packet transmission may be received by more than one visible satellite, resulting in more packets being successfully received than transmitted.

B. Energy budget evaluation

The energy budget is evaluated by considering the energy consumption associated with visible, collided, and Doppler-error packets. The energy E , in joules (J), required to transmit a single packet is calculated by

$$E = ToA \cdot P_{TX}, \quad (2)$$

where ToA is the packet Time-on-Air, expressed in seconds (s), and P_{TX} is the ED transmit power, expressed in watts (W). Knowing E and the number of packets in each category, the total energy consumption associated with each term can be calculated. With N_T , N_C , N_V , N_D , and N_S denoting the total number of transmitted, collided, visible, Doppler-failed, and successfully transmitted packets, respectively. The corresponding energy consumption, expressed in joules (J), is given by:

$$E_T = N_T \cdot E \quad (3)$$

$$E_C = N_C \cdot E \quad (4)$$

$$E_V = N_V \cdot E \quad (5)$$

$$E_D = N_D \cdot E \quad (6)$$

$$E_S = N_S \cdot E \quad (7)$$

Obtaining these values it is possible to compare them using bar charts to have an idea of how different configuration parameters impact in energy consumption.

IV. RESULTS

A. Test setups

We have evaluated two test setups which differs on bandwidth. The first batch of tests was conducted with $B = 125$ kHz, and the second batch with $B = 31.25$ kHz. For both it was chosen a constellation of 5 satellites:

- 1) STARLINK-1017
- 2) STARLINK-1012
- 3) STARLINK-1042
- 4) STARLINK-1048
- 5) STARLINK-1068

The number of EDs set to 20 with 100 packet per ED. The ED's position is shown in Fig. 1.

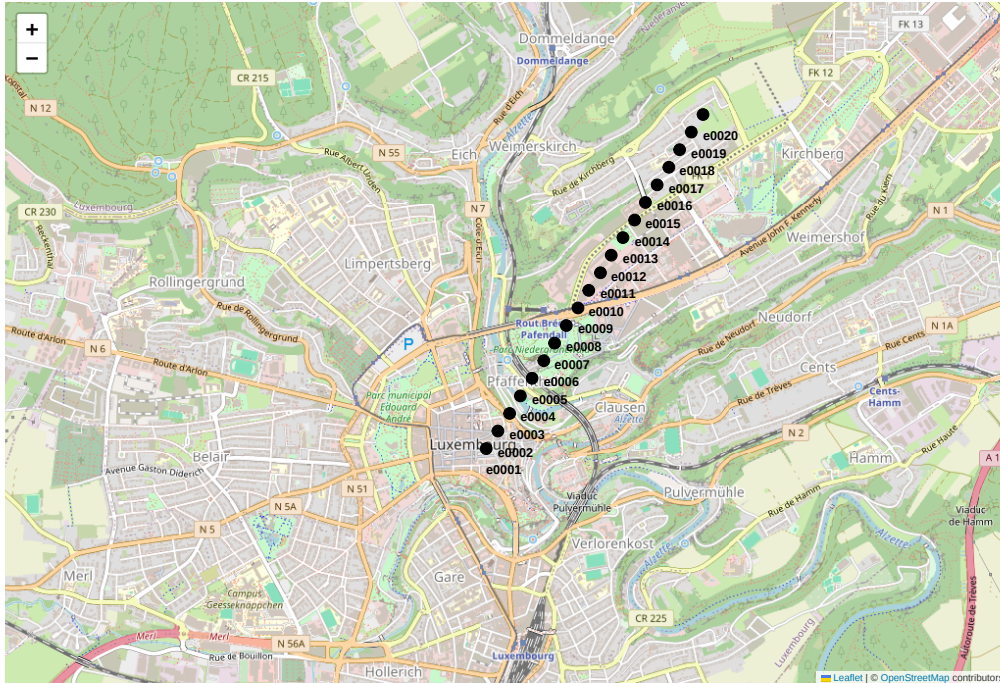


Fig. 1. End devices distribution.

The summary of the parameters used in both test batches is presented in the Table.I.

TABLE I
SUMMARY OF SIMULATION PARAMETERS

Category	Parameter	Value
Simulation	Initial time (t_i) UTC	2026-06-20 03:20:00
	Final time (t_f) UTC	2026-06-22 04:00:00
	Total duration	2 days and 40 min
Satellite	Number of satellites	5
	Minimum elevation angle ($^\circ$)	5
End Device (ED)	Number of EDs	20
	ED distribution	Line
	ED transmit power (W)	1
	ED transmission policy	Random
	Packets per ED	100
LoRa	Frequencies (MHz)	433, 868, 915
	Spreading Factor (SF)	7–12
	Bandwidth (kHz)	125, 31.25
	LDRO mode	OFF
	Packet length L (bytes)	35, 50, 100, 200

B. Results

The PDR evaluation, in addition to the PDR value, includes the collided, visible, Doppler error percentage of total transmissions, with one graph for each frequency= $\{433, 868, 915\}$ MHz and for each bandwidth $B=\{125, 31.25\}$ kHz. The energy budget evaluation was also calculated for both situations.

C. Frequency 433 MHz, B=125 kHz

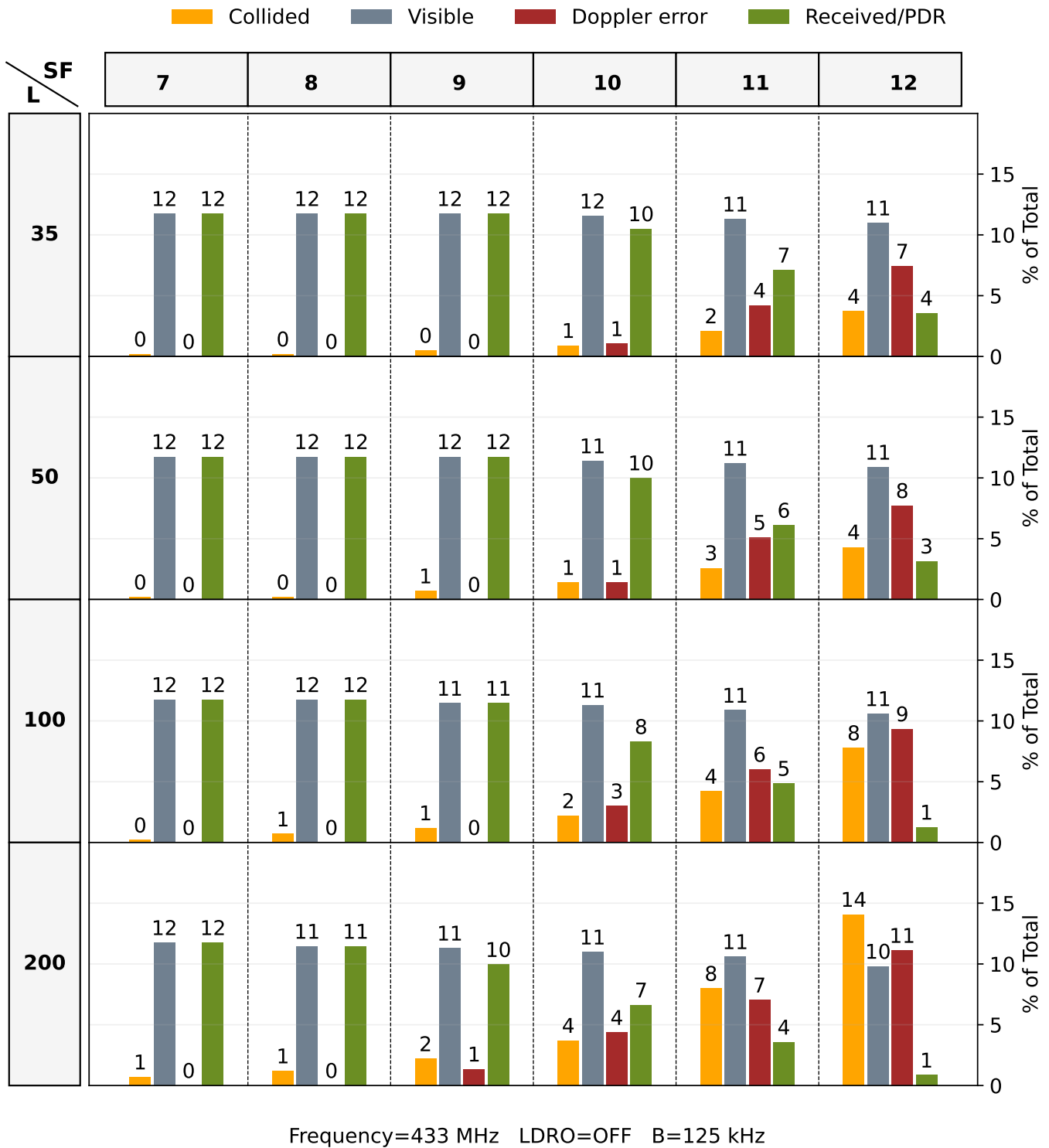


Fig. 2. Simulation config: frequency 433 Mhz, B=125kHz, 5 satellites, 20 ED with 100 packet sent per ED

D. Frequency 868 MHz, B=125 kHz

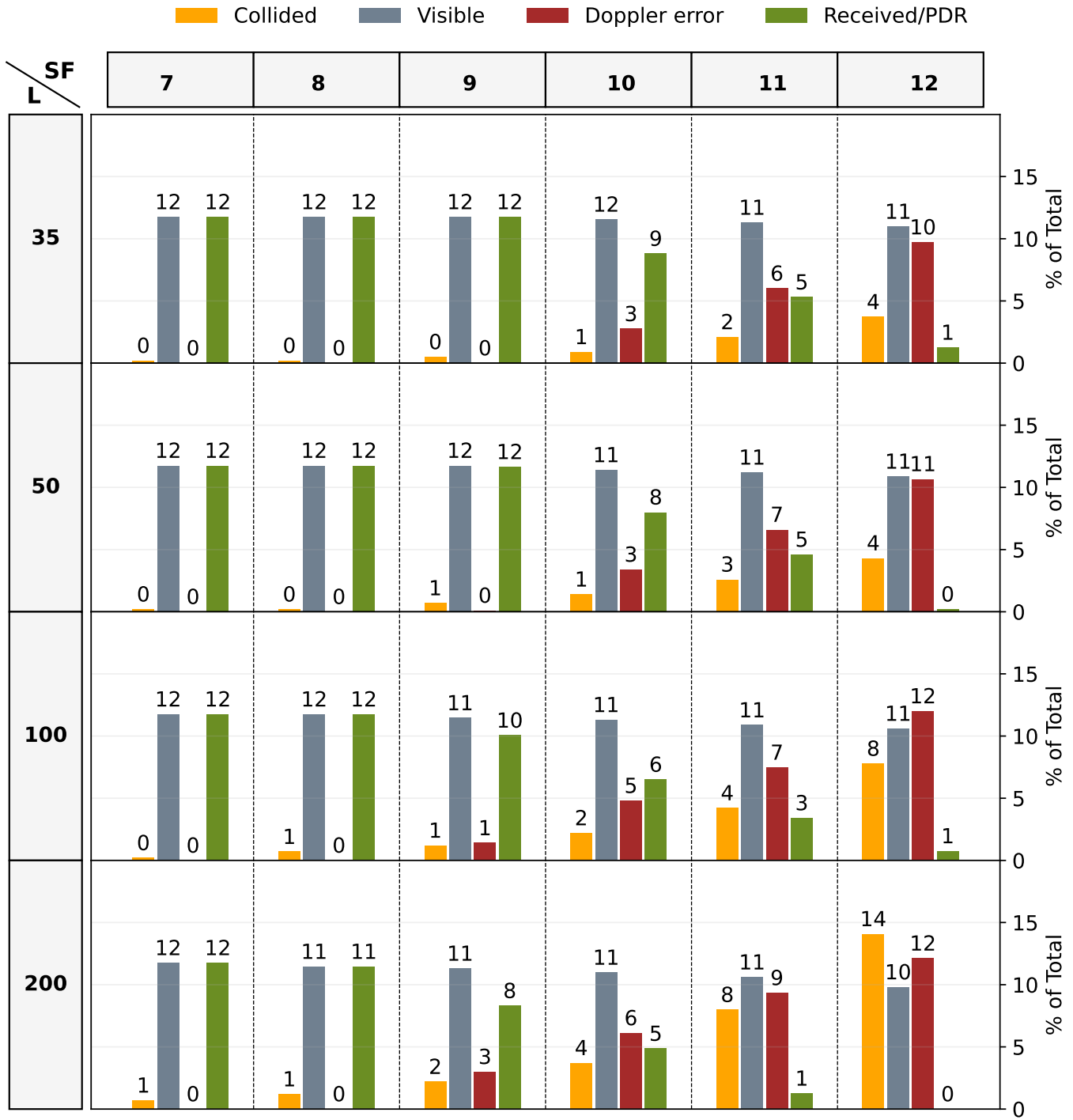


Fig. 3. Simulation config: frequency 868 Mhz, B=125 kHz, 5 satellites, 20 ED with 100 packet sent per ED

E. Frequency 915 MHz, B=125 kHz

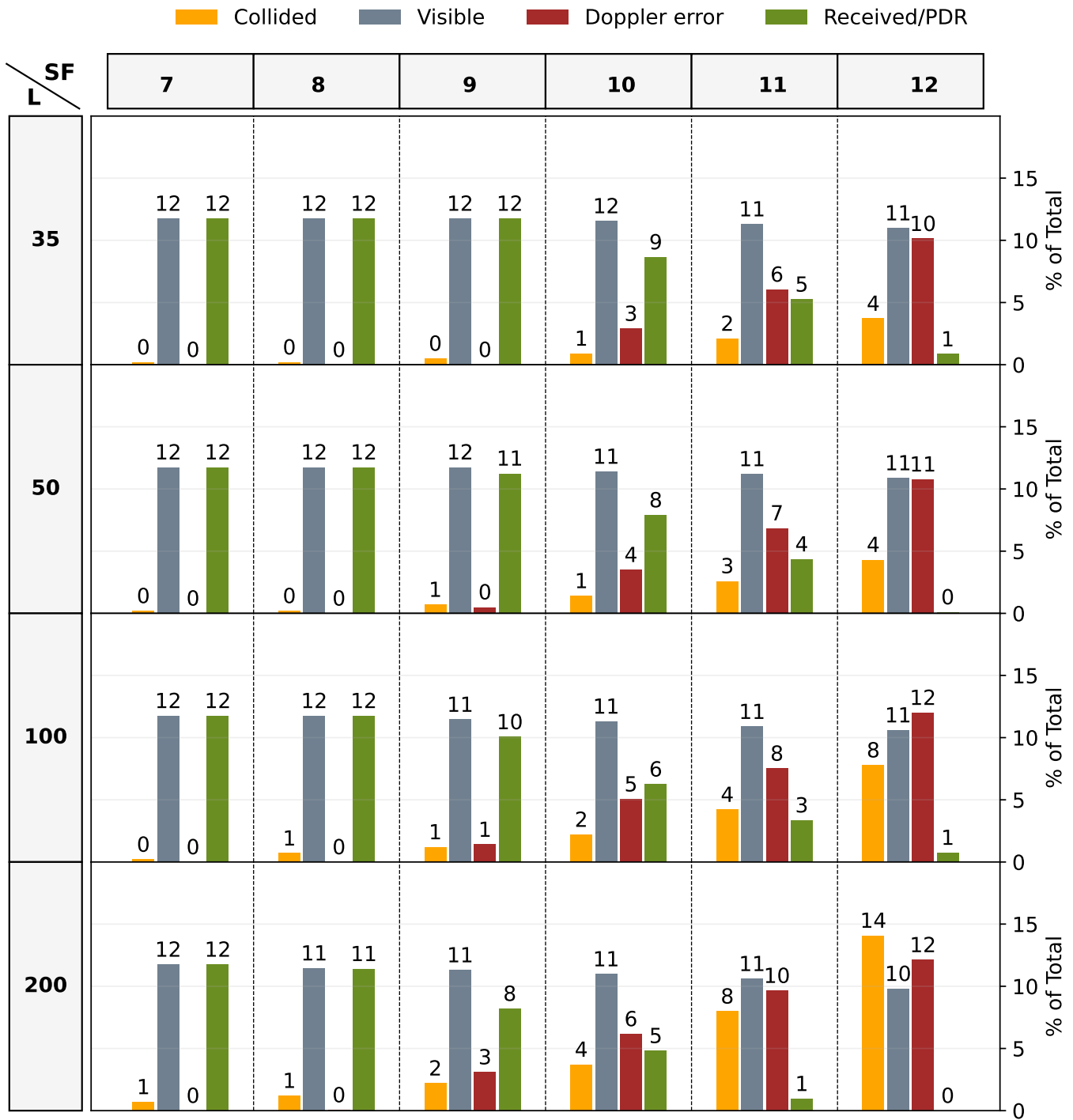
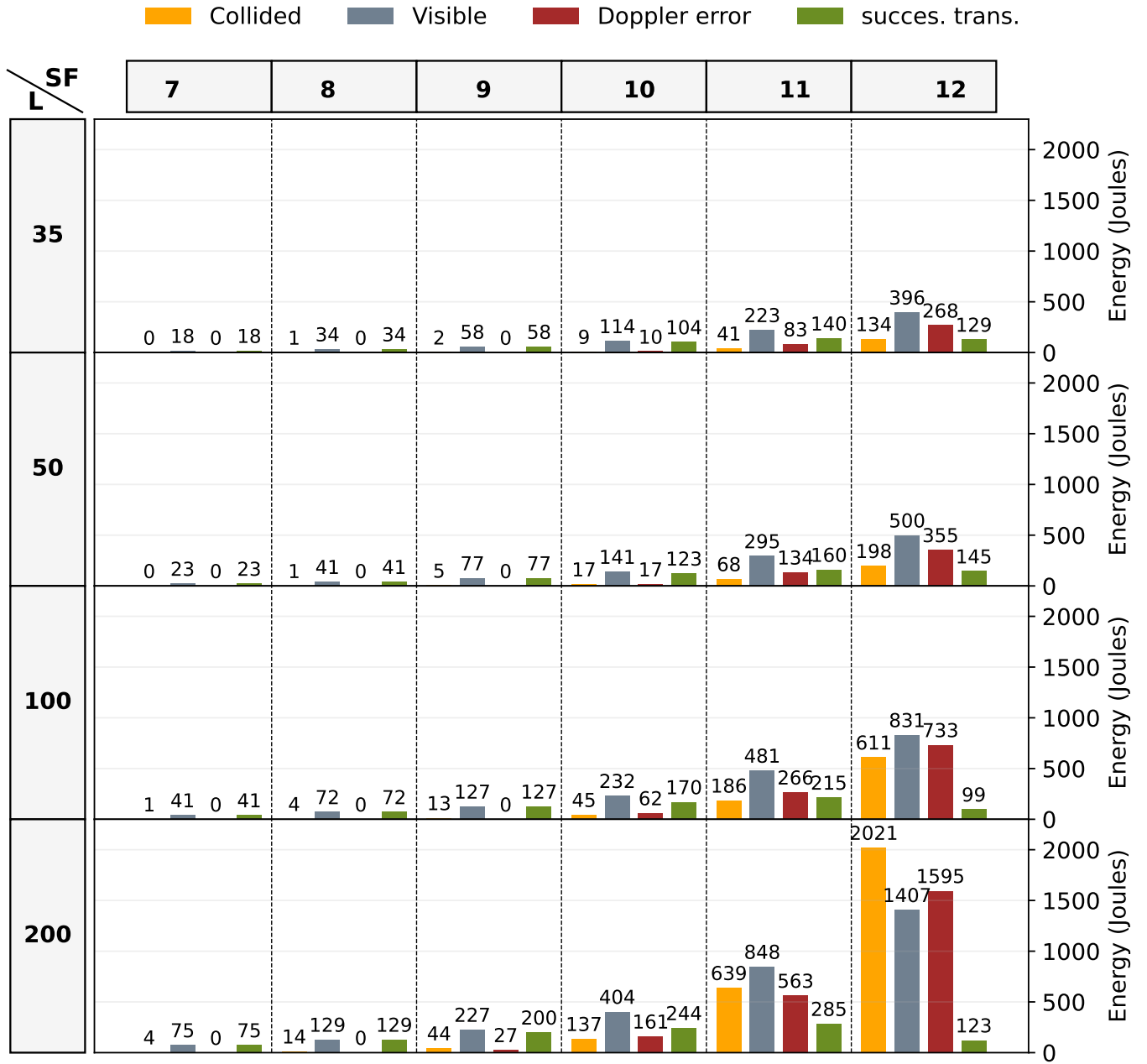


Fig. 4. Simulation config: frequency 915 Mhz, B=125 kHz, 5 satellites, 20 ED with 100 packet sent per ED

The Energy budget evaluation show the energy consumption of collided, visible, Doppler error and successfully transmissions.

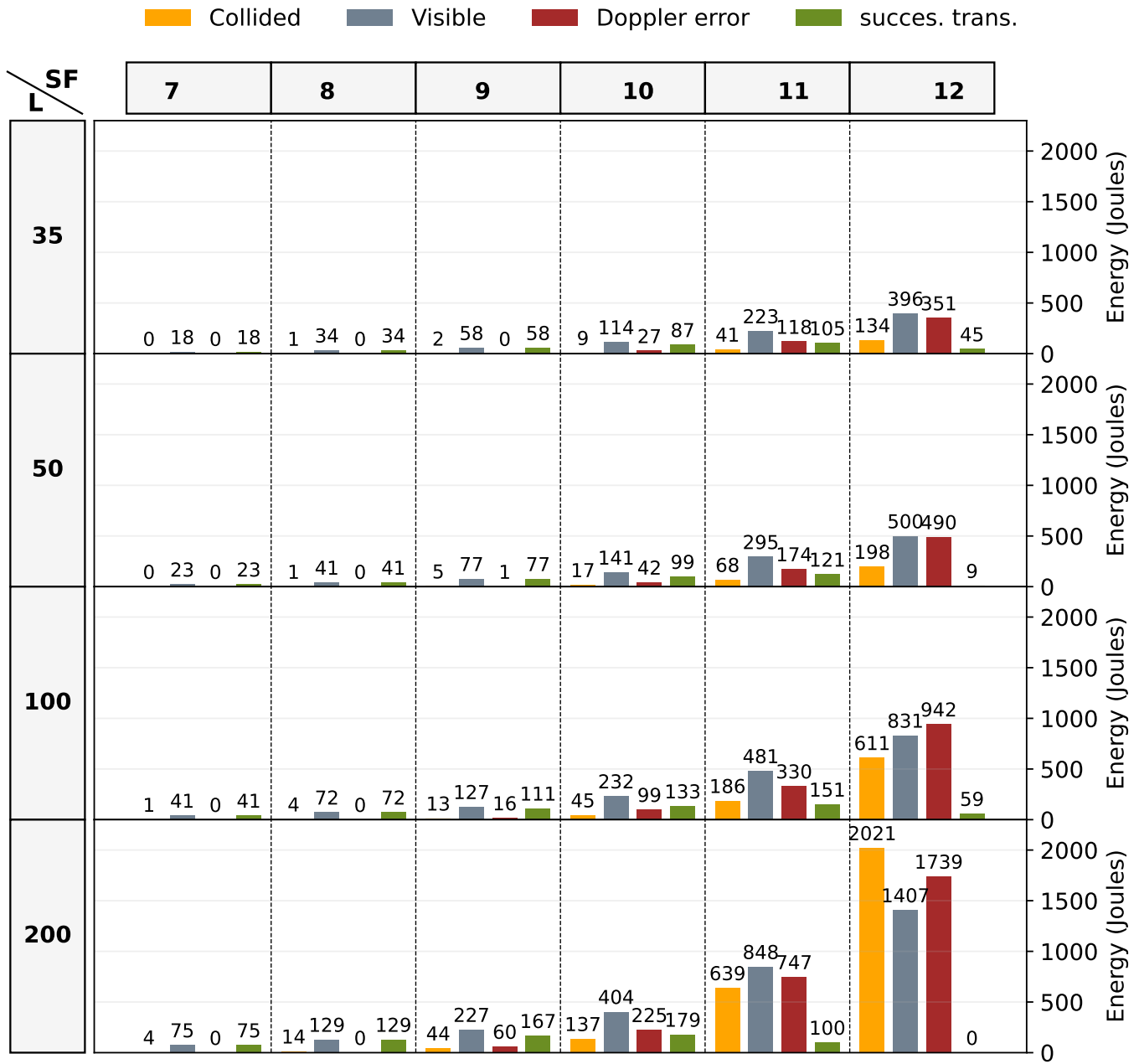
F. Frequency 433 MHz, B=125 kHz



Frequency=433 MHz LDRO=OFF B=125 kHz Tx power=1 W

Fig. 5. Simulation config: frequency 433 Mhz, B=125 kHz, 5 satellites, 20 ED with 100 packet sent per ED

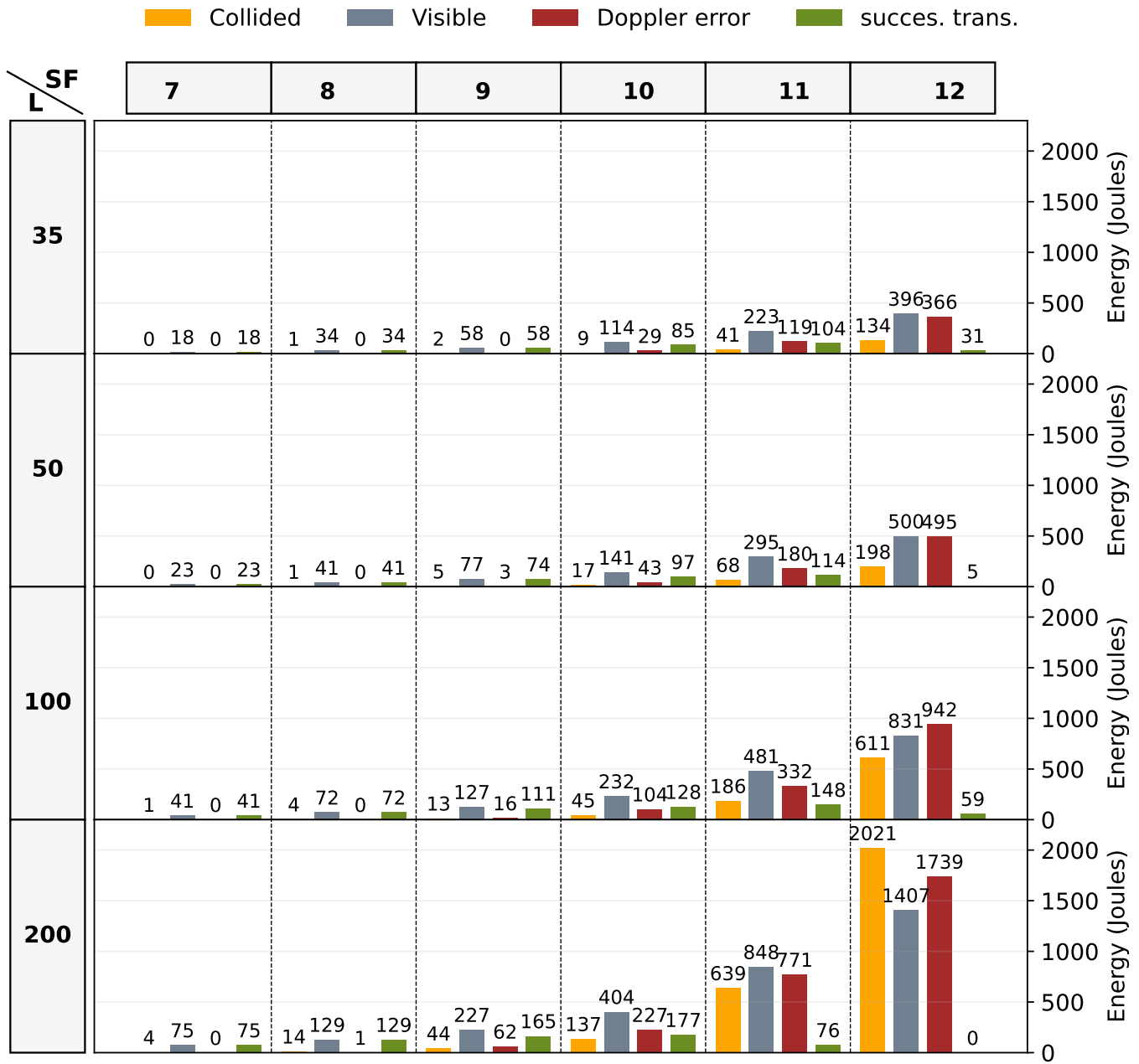
G. Frequency 868 MHz, B=125 kHz



Frequency=868 MHz LDRO=OFF B=125 kHz Tx power=1 W

Fig. 6. Simulation config: frequency 868 Mhz, B=125 kHz, 5 satellites, 20 ED with 100 packet sent per ED

H. Frequency 915 MHz, B=125 kHz



Frequency=915 MHz LDRO=OFF B=125 kHz Tx power=1 W

Fig. 7. Simulation config: frequency 915 Mhz, B=125 kHz, 5 satellites, 20 ED with 100 packet sent per ED

I. Frequency 433 MHz, B=31.25 kHz

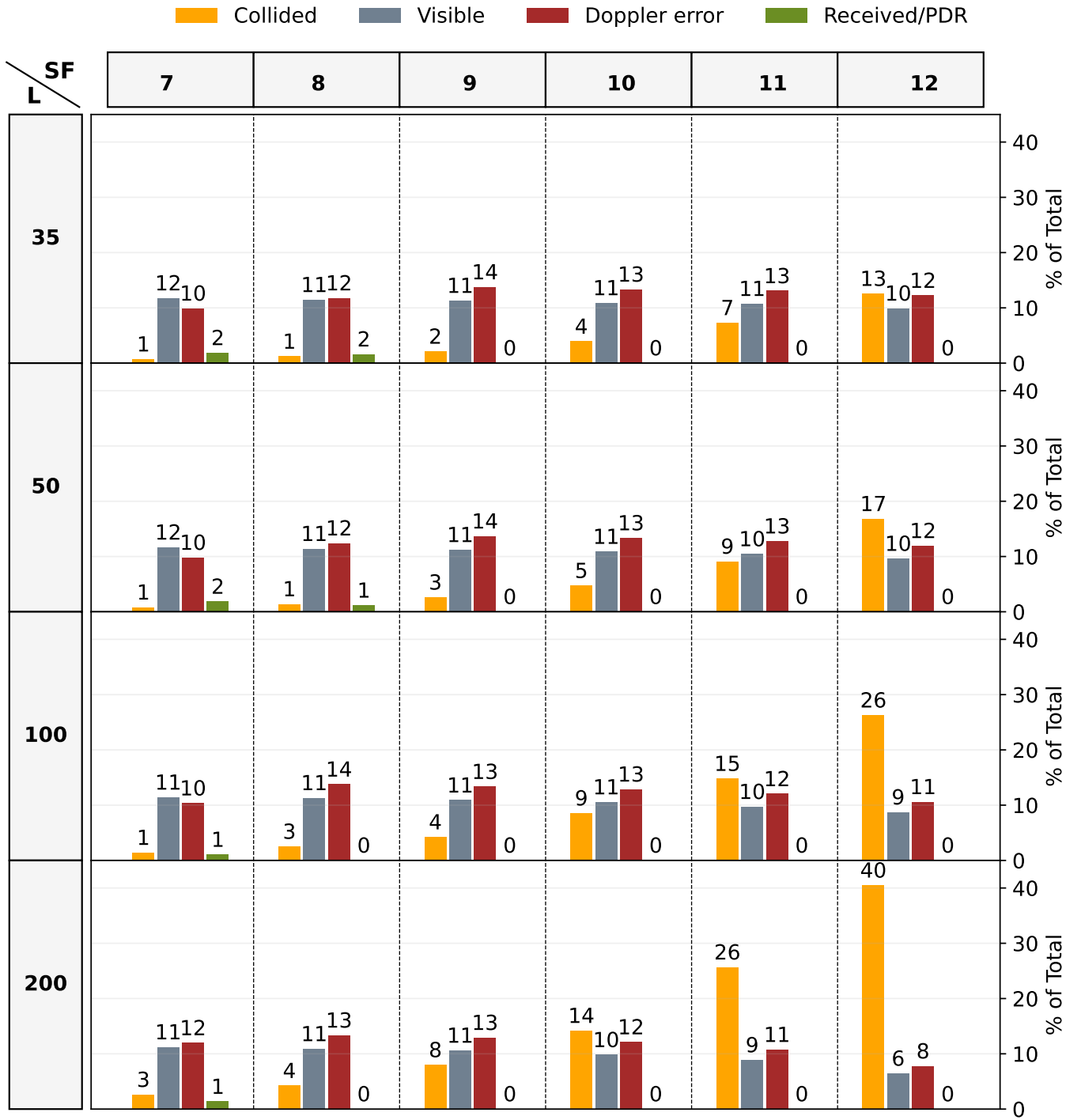


Fig. 8. Simulation config: frequency 433 Mhz, B=31.25kHz, 5 satellites, 20 ED with 100 packet sent per ED

J. Frequency 868 MHz, B=31.25 kHz

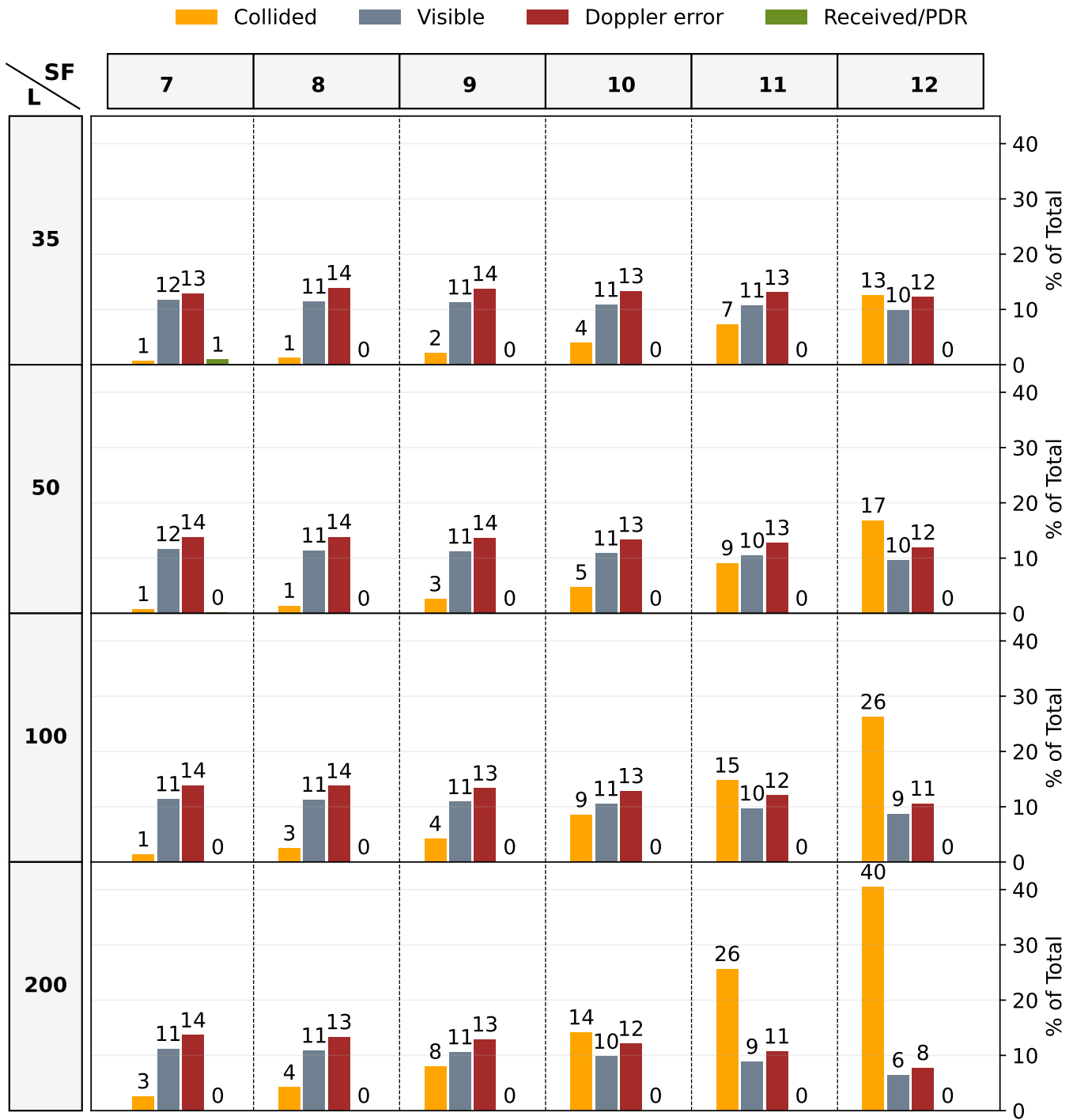


Fig. 9. Simulation config: frequency 868 Mhz, B=31.25 kHz, 5 satellites, 20 ED with 100 packet sent per ED

K. Frequency 915 MHz, B=31.25 kHz

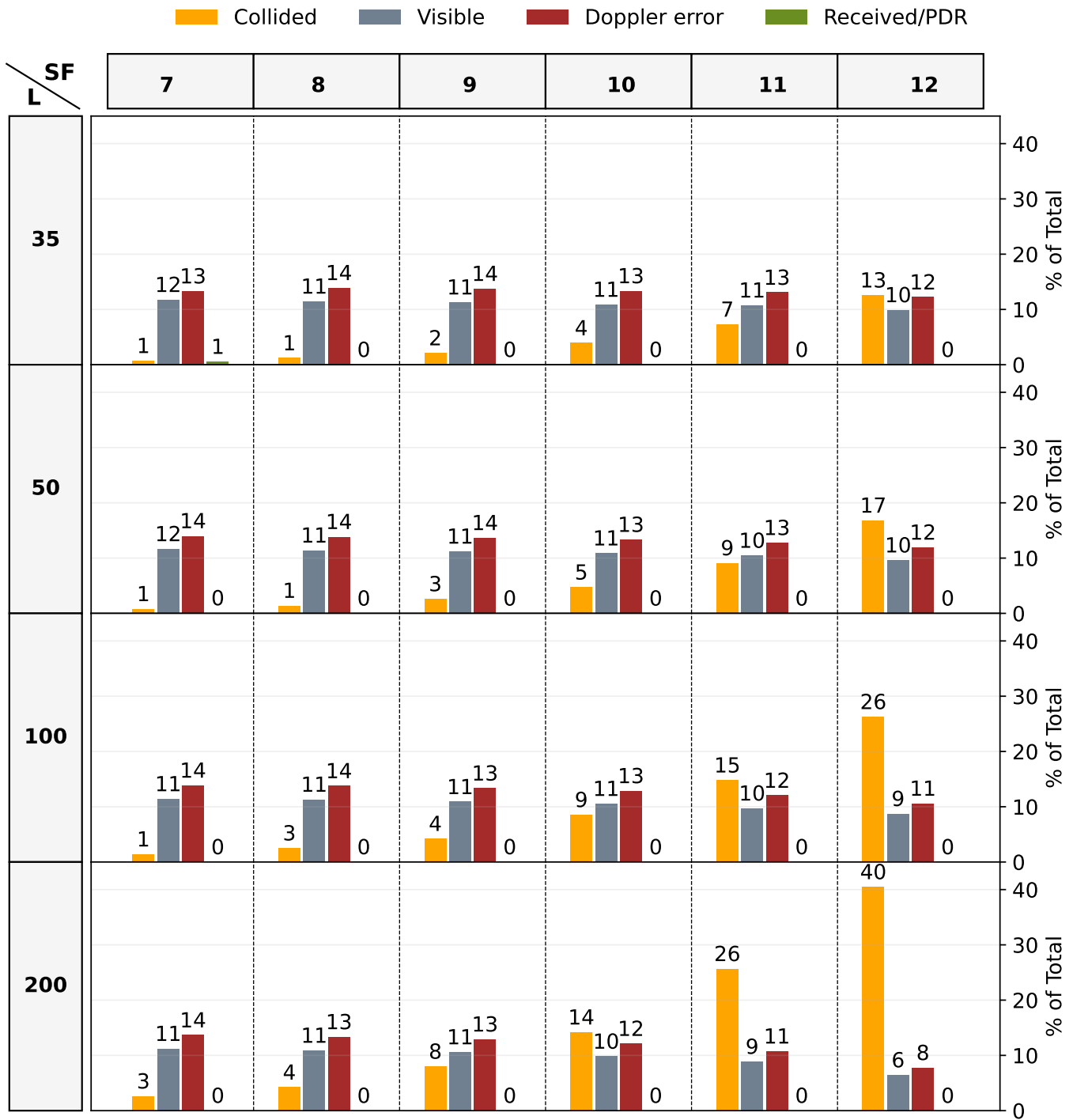


Fig. 10. Simulation config: frequency 915 Mhz, B=125 kHz, 5 satellites, 20 ED with 100 packet sent per ED

The Energy budget evaluation show the energy consumption of collided, visible, Doppler error and successfully transmissions.

L. Frequency 433 MHz, B=31.25 kHz

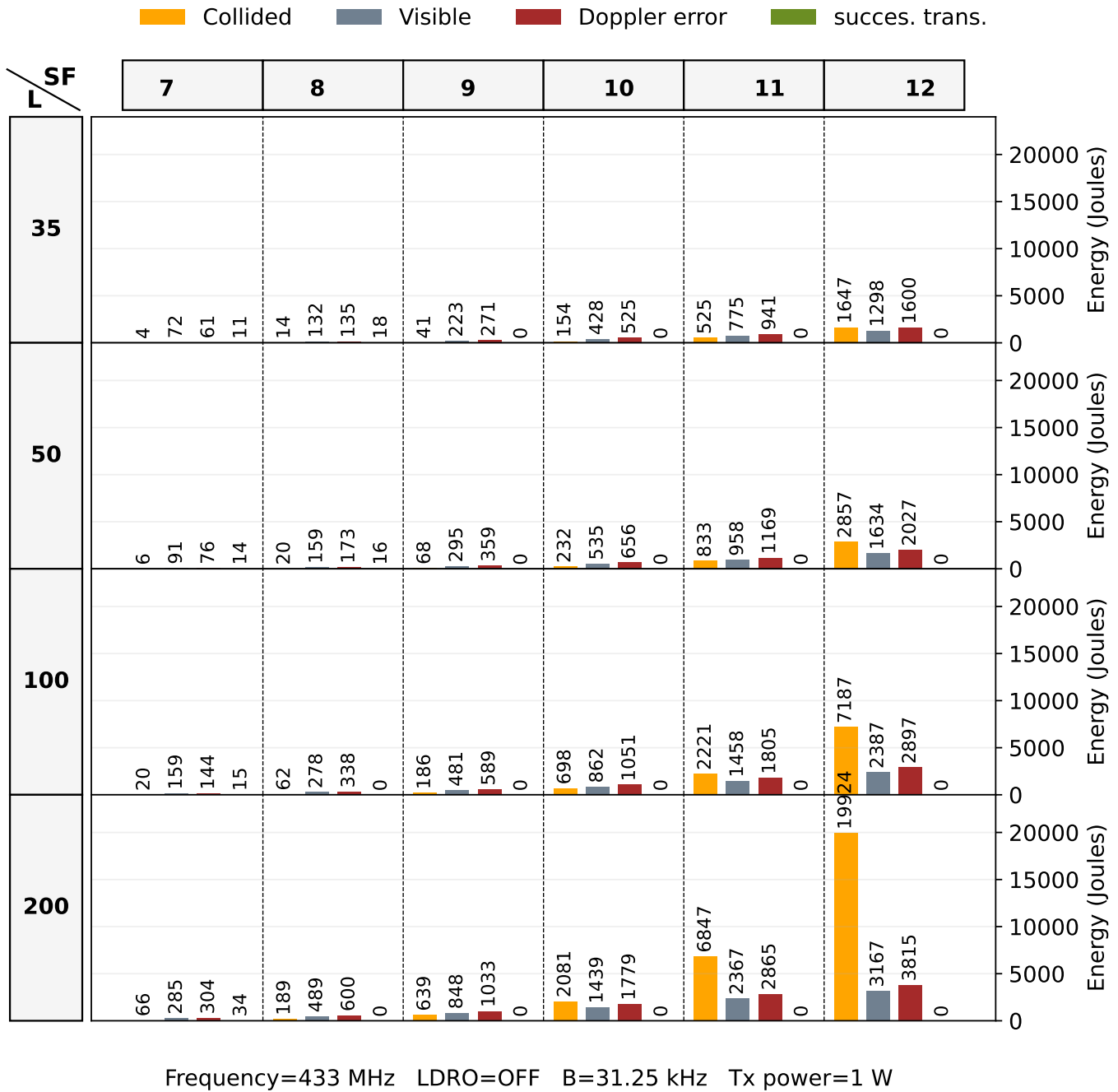


Fig. 11. Simulation config: frequency 433 Mhz, B=31.25 kHz, 5 satellites, 20 ED with 100 packet sent per ED

M. Frequency 868 MHz, B=31.25 kHz

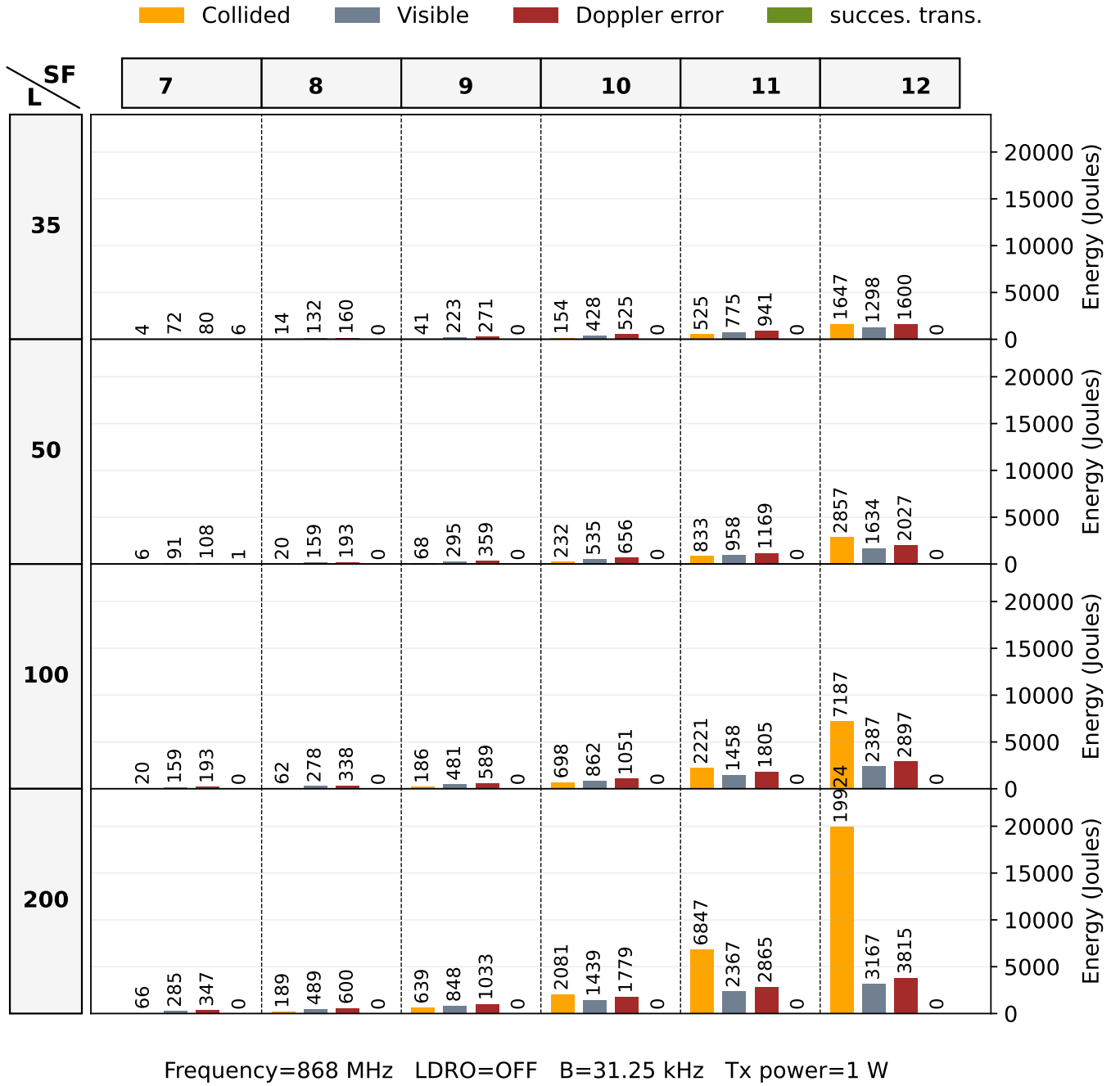


Fig. 12. Simulation config: frequency 868 Mhz, B=31.25 kHz, 5 satellites, 20 ED with 100 packet sent per ED

N. Frequency 915 MHz, B=31.25 kHz

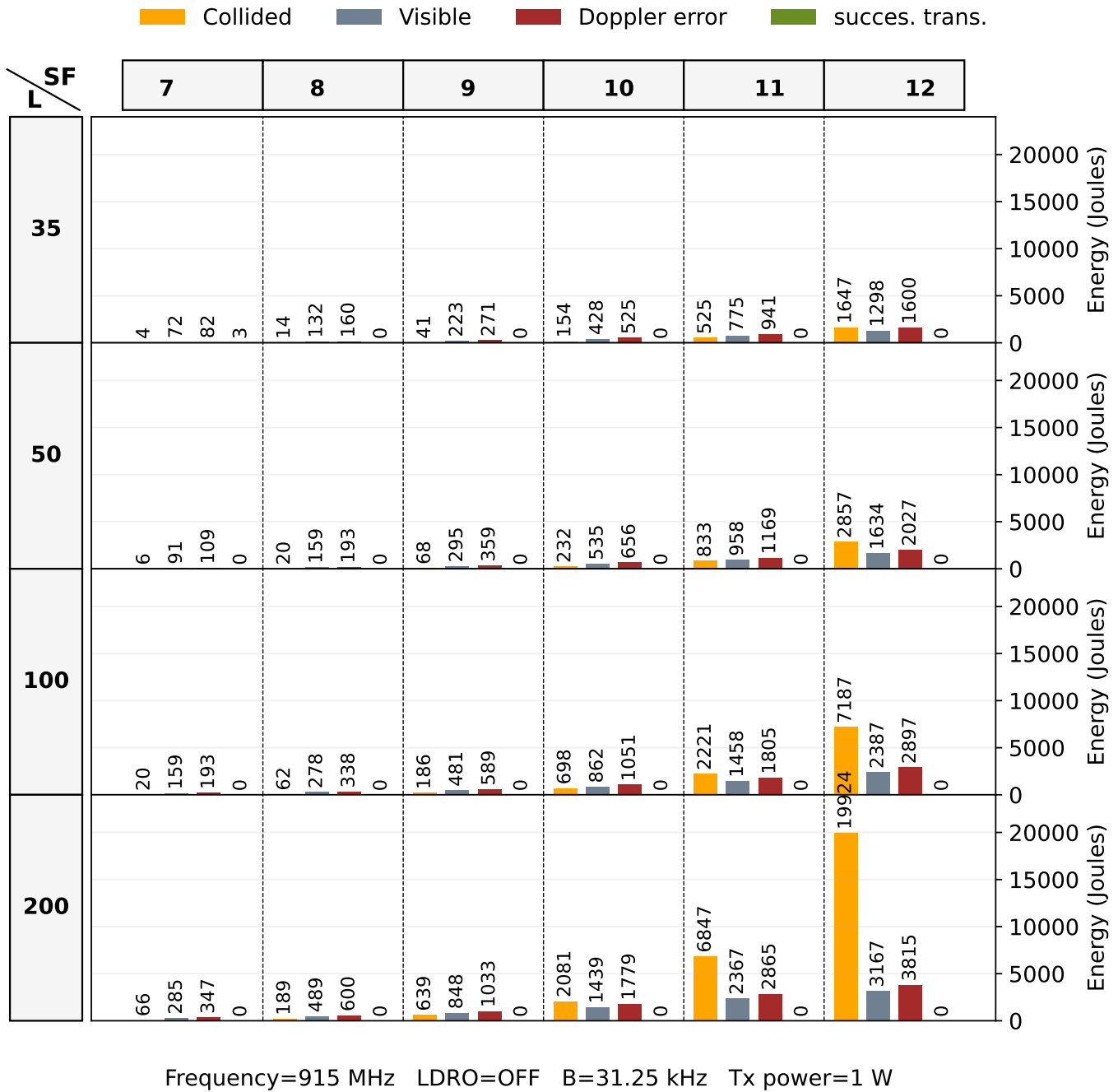


Fig. 13. Simulation config: frequency 915 Mhz, B=31.25 kHz, 5 satellites, 20 ED with 100 packet sent per ED

V. CONCLUSIONS

- **LDRO:** We need to review how LDRO is implemented in each code (LIST/CEDRA) to ensure that it is considered consistently in both implementations. We suggest creating three LDRO modes: "ON, OFF, and AUTO".
- **ED position:** We could discuss the possibility of creating additional options for distributing EDs. For example, EDs could be placed following a circular distribution within an area, randomly generated inside a circle, or using other positioning strategies.
- **ED transmission policy:** We could also discuss in more detail how the current transmission policy works and evaluate possible alternatives. One important observation is that, with the current policy, most packet losses are caused by non visibility issue.
- **Energy budget consideration:** The current energy budget analysis is not fully representative; it was performed only to provide an initial overview of how this parameter could affect the results. The analysis is not fair because different spreading factors (SFs) have different link budget limitations. We suggest adding a new evaluation parameter, in addition to visibility, collision, and Doppler effects: the link budget. This parameter would consider the total energy required for transmission, based on a complete link budget calculation using appropriate propagation and system models (we have one available if needed).
- **Data analysis:** We did not perform a numerical analysis of the data because this initial test was intended only as a starting point for discussion and further evaluation.